



Edible Coatings and Films for Seafood Preservation: Recent Advances and Future Perspectives

E Arivukumar¹, Mohamed Rizwan K² and Sneha A³

Department of Fish Quality Assurance and Management

Fisheries College and Research Institute,

Tamilnadu Dr.J.Jayalalithaa Fisheries University,

Tamilnadu- 628 008.

Abstract

Seafood is highly perishable due to its high moisture content, enzymatic activity, and susceptibility to microbial spoilage. Conventional preservation methods may affect the nutritional and sensory quality of seafood products. Edible coatings and films have emerged as eco-friendly and effective alternatives for extending the shelf life of seafood while maintaining product quality. These coatings act as protective barriers against moisture loss, oxygen penetration, and microbial contamination. This review discusses the applications of chitosan coatings, alginate films, antimicrobial edible coatings, their role in shelf-life extension, and commercial applications in the seafood industry.

Keywords: Seafood preservation, Edible coatings, Chitosan, Alginate films, antimicrobial coatings, Shelf-life extension.

Introduction:

Seafood is an important source of high-quality proteins, essential amino acids, vitamins, and omega-3 fatty acids. However, seafood products are highly susceptible to spoilage due to microbial growth, lipid oxidation, and enzymatic degradation. Traditional preservation methods such as refrigeration, freezing, and chemical preservatives are commonly used to maintain quality and safety. In recent years, edible coatings and films have gained considerable





attention as sustainable preservation technologies. These coatings form a thin protective layer on the food surface, reducing moisture loss, oxidation, and microbial contamination while maintaining the sensory and nutritional properties of seafood products.

2. Chitosan-Based Coatings

Chitosan is a natural biopolymer obtained through the deacetylation of chitin, which is abundantly found in crustacean shells such as shrimp and crab waste. It possesses excellent film-forming, biodegradable, and antimicrobial properties, making it one of the most widely studied edible coating materials. Chitosan coatings effectively inhibit the growth of spoilage microorganisms and foodborne pathogens in seafood products. Studies have shown that chitosan-coated fish fillets and shrimp exhibit reduced microbial counts, lower lipid oxidation, and improved sensory quality during refrigerated storage. The coating also acts as a barrier against oxygen and moisture, thereby extending product shelf life and maintaining freshness.

3. Alginate-Based Films

Alginate is a naturally occurring polysaccharide extracted from brown seaweeds. Due to its non-toxic, biodegradable, and film-forming properties, alginate has become an attractive material for edible film production. Alginate films create a semi-permeable barrier around seafood products, reducing moisture loss and delaying quality deterioration. These films can be combined with antioxidants and antimicrobial compounds to further enhance preservation efficiency. Alginate-based coatings have been successfully applied to fish fillets, shellfish, and ready-to-cook seafood products, helping to maintain texture, appearance, and freshness during storage.

4. Antimicrobial Edible Coatings

Antimicrobial edible coatings incorporate natural antimicrobial agents into edible film matrices to improve food safety and shelf-life extension. Common antimicrobial substances include essential oils, plant extracts, organic acids, and bioactive compounds derived from natural sources. These coatings inhibit the growth of spoilage microorganisms and pathogenic bacteria on seafood surfaces. Essential oils such as oregano, thyme, clove, and cinnamon oils have demonstrated strong antimicrobial activity when incorporated into edible coatings. The combination of natural antimicrobials with chitosan or alginate films provides enhanced protection against microbial contamination while reducing the need for synthetic preservatives.





5. Role in Shelf-Life Extension

Edible coatings and films play a significant role in extending the shelf life of seafood products. By forming a protective barrier around the product, they reduce oxygen transfer, moisture loss, and lipid oxidation. This helps maintain the texture, flavor, color, and nutritional quality of seafood during storage.

Furthermore, antimicrobial coatings suppress the growth of spoilage microorganisms, thereby delaying deterioration and improving food safety. Several studies have reported that coated seafood products exhibit longer storage stability and better sensory acceptance compared to untreated products. As a result, edible coatings represent an effective strategy for enhancing seafood quality and reducing post-harvest losses.





6. Commercial Applications

The commercial use of edible coatings and films in the seafood industry has increased significantly in recent years. Seafood processors are adopting these technologies to improve product quality, extend shelf life, and meet consumer demand for natural and sustainable food preservation methods. Chitosan-based coatings are widely used for fish fillets, shrimp, and shellfish products, while alginate films are commonly applied to fresh and frozen seafood. The incorporation of natural antimicrobial agents into edible coatings has further expanded their commercial potential. In addition to improving food safety, edible coatings contribute to reducing plastic packaging waste, supporting environmentally sustainable seafood processing practices.

7. Challenges and Future Prospects

Despite their numerous advantages, edible coatings and films face certain challenges, including production costs, scalability, and variations in coating performance under different storage conditions. Some coatings may also affect the appearance or texture of seafood products if not properly formulated. Future research should focus on developing multifunctional coatings with enhanced antimicrobial and antioxidant properties. Advances in nanotechnology, bioactive compounds, and smart packaging systems are expected to improve the effectiveness of edible coatings and expand their industrial applications. The growing emphasis on sustainable food processing and environmentally friendly packaging is likely to accelerate the adoption of edible coatings in the seafood industry.

8. Conclusion

Edible coatings and films have emerged as promising technologies for seafood preservation due to their ability to enhance food safety, extend shelf life, and maintain product quality. Chitosan coatings, alginate films, and antimicrobial edible coatings offer effective protection against microbial spoilage and quality deterioration. Although challenges remain regarding large-scale implementation, ongoing technological advancements are expected to strengthen their role in sustainable seafood processing. The increasing demand for natural preservation methods and eco-friendly packaging solutions highlights the significant future potential of edible coatings and films in the seafood industry.





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